

# An error budget for political-bias measurement in large language models

## Abstract

Large language models are widely reported to hold left-of-centre political views, a finding replicated across dozens of studies using standardized instruments. We show that this finding is substantially a property of the measurement. Administering two standardized instruments—the Political Compass and 8Values—to 19 models from 11 organizations, 60 times each (2,280 administrations), we decompose measured political position into variance attributable to model identity, to the instrument, and to residual trial noise. Instrument and axis choice accounts for 52% of variance and model identity for 13%: which questionnaire an auditor picks moves a reported score four times more than which model is audited, and trial-to-trial noise alone (35%) exceeds the model term. The instruments also fail convergent validity, with constructs they both claim to measure correlating at  $r = -0.26$  across models against  $r = +0.52$  for distinct constructs measured the same way. Published scores are therefore not comparable across instruments, and we propose a minimum reporting standard for such audits.

## 1 Introduction

Large language models now mediate political information for hundreds of millions of people, and a substantial literature reports that they lean left of centre. The finding has been replicated with standardized political instruments across model families, providers, and languages [1, 2, 3, 4, 5, 6, 7], and is now treated as an established property of the systems themselves.

Whether that property matters depends on whether it transfers to behaviour, and there is evidence that it does. Interacting with a politically biased model shifts the policy views and voting intentions of users who were undecided beforehand [8], models can be as persuasive as targeted human messaging on contested policy questions [9, 10], and merely *perceiving* a model as politically biased changes how much people trust its outputs on unrelated tasks [11]. A measured political lean is therefore a reasonable object of study, and auditing it is a reasonable thing to do.

The problem is the instrument. Political-bias audits of language models rest on questionnaires built for humans, administered in a single unattributed condition, and each of the assumptions that licenses this has since been challenged. Models accommodate the political identity they infer for the person asking, so an unattributed questionnaire partly measures sycophancy toward a presumed academic auditor [12]. The instruments themselves appear in pretraining data, so responses may reflect recognition of a known test rather than engagement with its items [13]. Forcing a model to pick among fixed options produces answers it does not give when allowed to respond freely [14], and the resulting scores have been argued not to measure the construct they name [15, 16]. Each of these has been demonstrated in isolation. None has been quantified against the others, and so

the field has no estimate of how much of a reported political score is the model and how much is the apparatus.

This paper supplies that estimate for one component of the apparatus. We administer two standardized instruments, the Political Compass and 8Values, to 19 models from 11 organizations, 60 times each, and decompose the resulting 2,280 measurements into variance attributable to model identity, to the instrument, and to residual trial noise. Instrument and axis choice accounts for 52% of variance and model identity for 13%: which questionnaire an auditor happens to choose moves a reported political score roughly four times more than which model is being audited, and trial-to-trial noise alone exceeds the model term. We then show that the two instruments fail the standard psychometric test of convergent validity, so their scores are not interchangeable even in principle. We do not claim models have no political lean. We claim that the published magnitudes are not comparable across studies that used different instruments, and that an error budget should accompany any such audit.

## 2 Results

### 2.1 The standard design reproduces the standard finding

Administered the way the field administers them—one unattributed asker, fixed response options, 60 repetitions per model per instrument—both instruments place all 19 models in the same quadrant. Every model scores economically left and socially libertarian on the Political Compass, and toward equality, peace, liberty and progress on 8Values (Fig. 1). Classical ANOVA, Welch’s ANOVA and Kruskal–Wallis agree at  $p < .0001$  on every axis, with large effect sizes ( $\eta^2 = 0.56\text{--}0.73$ ), and the magnitude of the lean varies substantially between models. This is what the published literature reports, and we reproduce it. We present it here not as our finding but as the object of measurement: what the field’s method yields when applied carefully to a broad model set.

Political Compass position of all 19 models (mean of 60 trials each); + marks the neutral human-baseline proxy

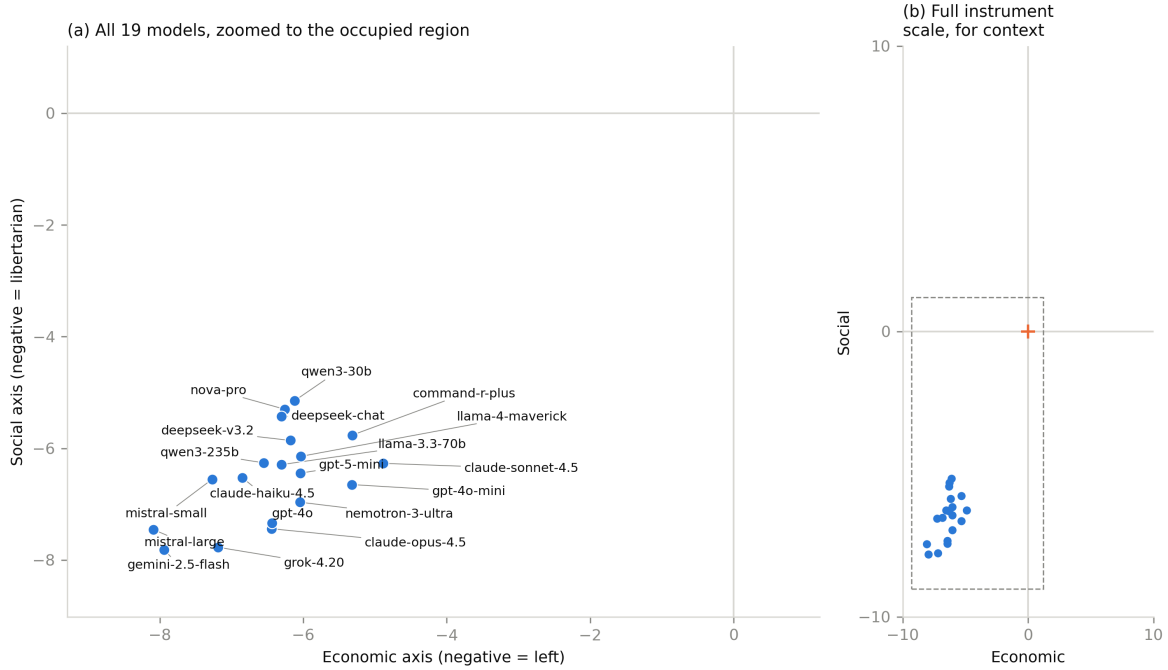


Figure 1: Baseline replication under the standard single-condition design. Mean Political Compass position of all 19 models, 60 administrations each. All models fall in the economically left, socially libertarian quadrant, reproducing the published finding.

The cohort is also narrow in absolute terms. Across the six axes, the full spread between the most and least extreme model occupies between 13.0% and 22.2% of each instrument’s response range. The headline claim is not that models sit at an extreme, but that they sit close together.

## 2.2 The error budget

Decomposing the 2,280 administrations into crossed random effects (Fig. 2, Table 1) gives a different picture of what those scores contain. Pooled across instruments, the choice of instrument and axis accounts for 52.2% of variance in a measured political position. Model identity accounts for 13.0%. Residual trial-to-trial noise—the same model, the same instrument, the same prompt, a different repetition—accounts for 34.8%, more than twice the model term. The model-by-axis interaction is negligible (0.04%), meaning models do not reorder themselves between axes; they are simply spread much less than the instruments are.

The error budget: how much of a measured political position is the model?

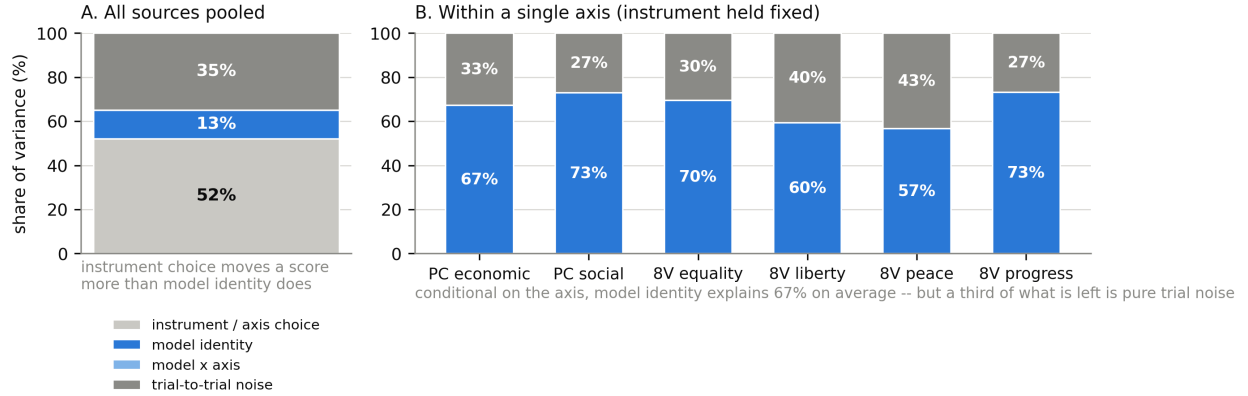


Figure 2: **The error budget.** Variance components of a measured political position, from a crossed random-effects model on 2,280 administrations. (a) Pooled across instruments: instrument and axis choice accounts for 52% of variance, model identity for 13%, and residual trial-to-trial noise for 35%. (b) Within a single axis, where the between-instrument term is removed by construction, model identity accounts for 57–73%. The two panels answer different questions: panel (a) governs comparison *across* studies using different instruments, panel (b) governs comparison *within* one study. Quoting either alone is misleading.

Table 1: Error budget. Variance components from a crossed random-effects model on the 2,280 administrations. **Pooled over all axes**, instrument and axis choice accounts for 52% of variance and model identity for only 13%. The per-axis rows below are *conditional on a fixed axis*, which removes the between-axis term and is why the model share is much larger there. The two denominators answer different questions and must not be quoted interchangeably.

| Axis        | Model identity (%) | Trial noise (%) | <i>n</i> |
|-------------|--------------------|-----------------|----------|
| PC economic | 67.3               | 32.7            | 1140     |
| PC social   | 73.0               | 27.0            | 1140     |
| 8V equality | 69.6               | 30.4            | 1140     |
| 8V liberty  | 59.5               | 40.5            | 1140     |
| 8V peace    | 56.9               | 43.1            | 1140     |
| 8V progress | 73.2               | 26.8            | 1140     |
| <b>Mean</b> | <b>66.6</b>        | <b>33.4</b>     |          |

The practical reading is that which questionnaire an auditor happens to pick moves a reported political score about four times more than which model is being audited, and that repeating the same measurement moves it more than twice as much. A score reported without an instrument and a repetition count is therefore not interpretable, and two studies using different instruments are not measuring on a common scale.

This does not mean model identity is undetectable. Conditional on a fixed axis, where the between-instrument term is removed by construction, model identity accounts for 66.6% of the remaining variance, ranging from 56.9% on 8Values peace to 73.2% on 8Values progress (Fig. 2b). Models

are clearly separable once the instrument is held still. Both numbers are correct and they answer different questions; the 13% figure is the one that governs comparisons across studies, and the 66.6% figure is the one that governs comparisons within a single study. Reporting either alone is misleading, which is why we report the decomposition rather than a single coefficient.

### 2.3 The two instruments are not measuring the same constructs

If instrument choice moves scores this much, the natural defence is that the instruments are noisy measures of one underlying political position. A multi-trait multi-method analysis rejects this. Treating the economic and social constructs as traits and the two questionnaires as methods, convergent validity—the same construct measured two ways—averages  $r = -0.26$  across the 19 models, with the social construct at  $r = -0.51$ . Discriminant validity—different constructs measured the same way—averages  $r = +0.52$ . The Campbell–Fiske criterion requires the first to exceed the second; here the ordering is inverted, and the convergent coefficient is negative (Fig. 3, Table 2).

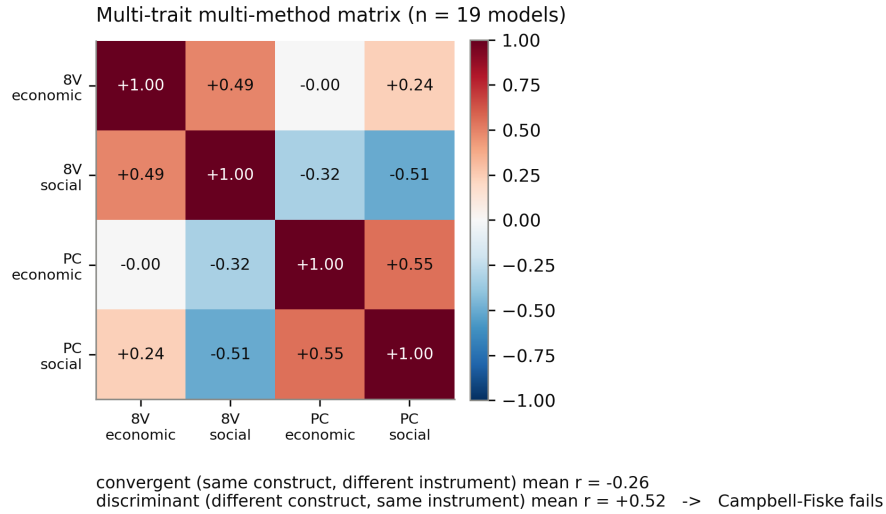


Figure 3: Multi-trait multi-method matrix across the 19 models. Convergent coefficients (same construct, different instrument) average  $r = -0.26$ ; discriminant coefficients (different construct, same instrument) average  $r = +0.52$ . The Campbell–Fiske ordering is inverted, so the two instruments cannot be treated as interchangeable measures of a common political position.

Table 2: Multi-trait multi-method coefficients across the 19 models. Convergent validity requires that the same construct measured by two instruments correlate more strongly than two different constructs measured by the same instrument. It does not.

| Type                     | Pair                                                   | $r$           |
|--------------------------|--------------------------------------------------------|---------------|
| Convergent               | 8values:economic vs political_compass:economic         | -0.005        |
| Convergent               | 8values:social vs political_compass:social             | -0.506        |
| Discriminant             | 8values:economic vs 8values:social                     | +0.490        |
| Discriminant             | political_compass:economic vs political_compass:social | +0.549        |
| <b>Mean convergent</b>   |                                                        | <b>-0.255</b> |
| <b>Mean discriminant</b> |                                                        | <b>+0.520</b> |

A model ranked as economically left by one instrument is, if anything, ranked less so by the other. Separately, the four 8Values axes correlate with each other at mean  $r = 0.71$  (liberty-peace,  $r = 0.85$ ), which is the signature of one dominant general factor rather than four separable traits. Taken together, these results mean the instruments cannot be treated as interchangeable measures of a common quantity, and that scores from studies using different instruments should not be pooled or compared numerically.

## 2.4 Stability is item-specific, and the published stability metric is invalid

Aggregating consistency to the axis level hides where inconsistency actually lives. Measured per item, 44.8% of model-item cells are perfectly stable across all 60 repetitions, and the most volatile decile of items carries only 14.4% (8Values) and 15.9% (Political Compass) of total response entropy (Fig. 4). Instability is a property of particular contested items, not a global trait of a model, and a model-level stability score averages the two together.

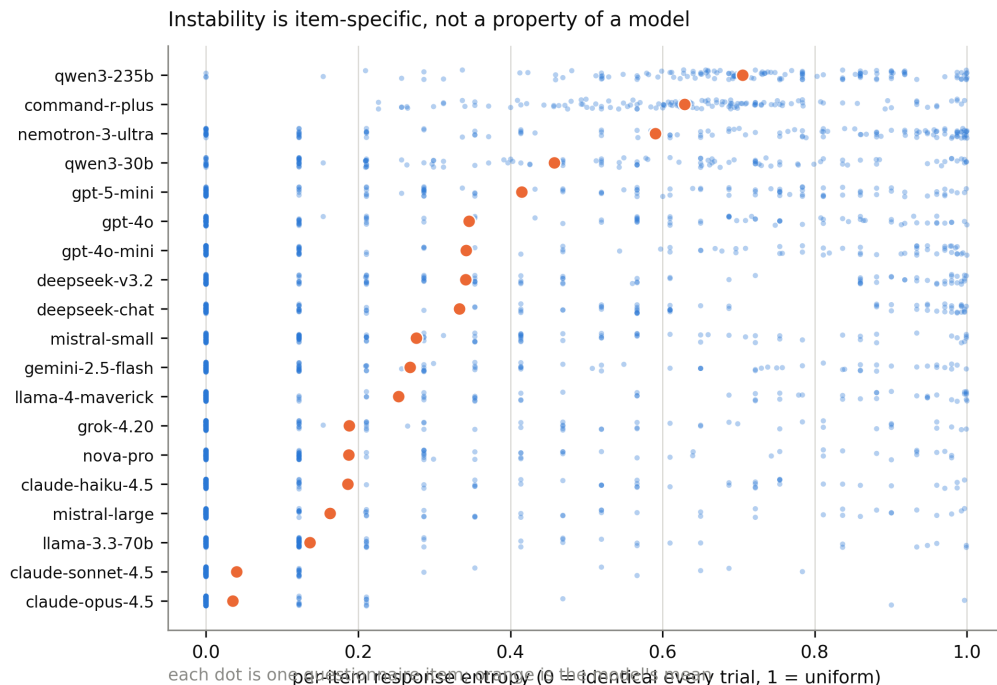


Figure 4: Per-item response entropy by model. Each point is one questionnaire item; orange marks the model mean. 44.8% of model-item cells are perfectly stable across all 60 repetitions, and the most volatile decile of items carries under 16% of total entropy, so inconsistency is concentrated in particular contested items rather than distributed across a model’s responses.

The stability metric used in this literature, including in an earlier version of this manuscript, compounds the problem. Normalising dispersion by each sample’s own observed range makes the statistic scale-free in the wrong way: a model whose 60 answers fall in a 0.12-wide interval divides by 0.12. The most extreme case here is claude-opus-4.5 on the Political Compass economic axis, which registers a stability score of 65.2—among the worst in the cohort—on a standard deviation of 0.060, which is among the best (Fig. 5). Across all 114 model-axis cells the published

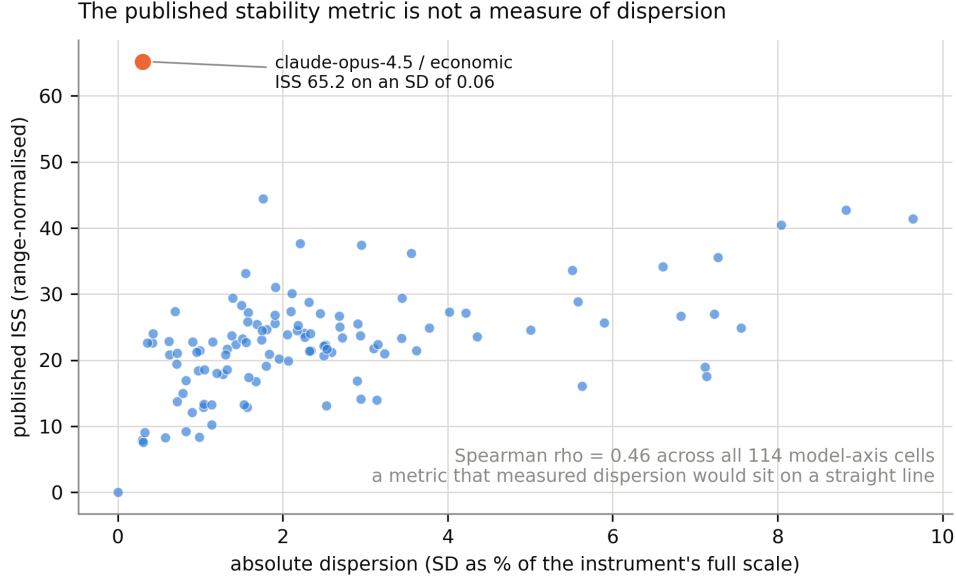


Figure 5: The range-normalised stability metric is not a measure of dispersion. Each point is one model-axis cell. Normalising by each sample’s own observed range inflates the score for models whose answers are tightly clustered: the highlighted cell registers 65.2 on a standard deviation of 0.060. Across all 114 cells the metric correlates with absolute dispersion at only  $\rho = 0.47$ .

metric correlates with absolute dispersion at only  $\rho = 0.47$ . We therefore report dispersion in the instrument’s own units, and the residual variance component of the error budget, in its place.

## 2.5 What this leaves open

The decomposition above quantifies one component of the measurement apparatus: the instrument. Three further components have been demonstrated in the literature but are not estimated here—accommodation to the inferred asker [12], recognition of contaminated instruments [13], and forced-choice framing [14]—and each would enter this budget as an additional stratum. The design required to estimate them alongside the instrument term is specified in the Methods.

## 3 Discussion

The finding that language models lean left of centre is real in the sense that it reproduces: we obtain it here on 19 models and two instruments, as others have obtained it elsewhere. What we add is a denominator. When the measured quantity is decomposed, the instrument contributes four times more variance than the model, and simple repetition of the same measurement contributes more than twice as much. The lean survives, but the *magnitude* of a reported lean is mostly not about the model.

That distinction matters because the literature’s quantitative claims are magnitude claims. Statements that one model is more partisan than another, that a provider’s alignment procedure shifts a model’s politics by some amount, or that lean has grown or shrunk across model generations, all

require that scores be comparable across studies. Our multi-trait multi-method result says they are not: the two most widely used instruments produce negatively correlated rankings of the same construct across the same models. Pooling or contrasting scores obtained with different instruments is therefore not a conservative approximation; it is an operation with no defined meaning.

We want to be precise about the scope of this claim, because it is easy to overstate. We are not arguing that political-bias audits are uninformative, nor that the models are politically neutral, nor that any particular published paper is wrong. Within a single study using a single instrument, model identity is clearly measurable and accounts for roughly two-thirds of the remaining variance; rank orderings obtained that way are meaningful. The failure is one of transportability. It also has a straightforward remedy, which is to report the apparatus alongside the estimate rather than the estimate alone.

Our own earlier analysis is among the work this corrects. The stability metric we previously reported normalised dispersion by each sample’s own range, which assigns one of the worst stability scores in the cohort to a model whose answers varied by six hundredths of a point. We report that here rather than quietly replacing it, because the failure mode is not unique to us: any scale-free consistency statistic that divides by observed spread will invert in the same way on tightly clustered responses, and several published consistency measures are constructed this way.

Three components of the apparatus remain unquantified in this design. Accommodation to the inferred asker [12], recognition of instruments present in pretraining data [13], and the constraint imposed by forced-choice response formats [14] have each been demonstrated in isolation, and each would enter the budget in Fig. 2 as a further stratum. The factorial that estimates all of them jointly with the instrument term is specified in the Methods; until it is run, the 13% attributable here to model identity should be read as an upper bound, since every additional measurement factor can only take variance away from it. That is the direction of the remaining uncertainty, and it does not favour the published interpretation.

### Box 1: A minimum reporting standard for political-bias audits

On the evidence above, we suggest that a political-bias audit of a language model should report the following alongside any score. None of these is expensive; together they make an estimate interpretable and comparable.

1. **The instrument, named, with its items.** Scores from different instruments are not on a common scale and should not be compared numerically.
2. **The asker condition.** State the framing presented to the model, including the absence of one. An unattributed prompt is a condition, not a neutral baseline.
3. **Repetition count and dispersion.** Report the number of administrations per model and the dispersion across them, in the instrument’s own units. Trial noise here exceeded the model effect.
4. **An error budget, or at minimum a variance decomposition** across whatever measurement factors were varied. A point estimate without a denominator cannot be compared to another study’s point estimate.

5. **A contamination check** for instruments that appear in pretraining corpora, such as an item-inversion or paraphrase control.
6. **Item-level results**, or at least the dispersion across items. Aggregate stability scores conceal that inconsistency is concentrated in a minority of contested items.

Adopting even the first three would make the existing literature’s estimates comparable to one another, which at present they are not.

## 4 Methods

### 4.1 Research Design

This study employs a quantitative, comparative design: every model is administered the same two standardized instruments under the same protocol, so that bias magnitude and trial-to-trial consistency can be compared directly across models rather than assessed qualitatively per model. This follows a design paradigm sometimes called “machine psychology” [17], administering instruments designed for human respondents directly to LLMs and analyzing the results with the same psychometric tools used for human data. Treating an LLM’s answers to a human-designed instrument as meaningful, comparable data is not self-evident and rests on the notion of *algorithmic fidelity*: that, under the right conditions, language-model outputs can approximate the response patterns of the human populations reflected in their training and tuning data closely enough to support this kind of comparative analysis [18]. Directly administering standardized psychometric instruments to language models and scoring the results as if from a human respondent has precedent both for personality and demographic inventories [19] and, at larger scale, for a resampling design close to this study’s own repeated-trials approach [20]. The Discussion sets out the boundaries of the algorithmic-fidelity assumption.

### 4.2 Model Roster

Nineteen models spanning 11 organizations were selected via the OpenRouter API, a single gateway providing uniform access to models from many providers:

OpenAI [21]: gpt-4o-mini, gpt-5-mini, gpt-4o  
 Anthropic [22]: claude-haiku-4.5, claude-sonnet-4.5, claude-opus-4.5  
 DeepSeek [23]: deepseek-chat, deepseek-v3.2  
 Google: gemini-2.5-flash  
 Meta: llama-3.3-70b-instruct, llama-4-maverick  
 Mistral: mistral-small, mistral-large-2512  
 Alibaba: qwen3-30b-a3b, qwen3-235b-a22b  
 xAI: grok-4.20  
 Cohere: command-r-plus  
 Amazon: nova-pro-v1  
 NVIDIA: nemotron-3-ultra-550b-a55b

The roster spans both closed and open-weight models, four countries (United States, China, France, and Canada), and roughly two orders of magnitude in per-token price, chosen to approximate the

model diversity of comparable published studies rather than relying on a small, convenience-sampled set of flagship models; 19 models matches the largest LLM roster used by a directly comparable recent study on this exact topic [5].

One additional model, `google/gemini-2.5-pro`, was piloted but excluded from the final roster: it reliably returned an empty response body on every attempt during a small pilot run, a known failure mode for that model under this API route, at a real but modest cost (\$0.12 across four failed attempts) before the decision was made to drop it. Google remains represented in the final roster via `gemini-2.5-flash`.

### 4.3 Political Typology Test Selection

Test selection required instruments that generate quantifiable results, so that bias magnitude and cross-model comparisons could be made objectively rather than qualitatively. We selected the Political Compass and 8Values [24, 25] for their clear, structured question formats. Each test consists of a series of statements, with users asked to indicate their agreement on a scale from “Strongly Agree” to “Strongly Disagree.” The Political Compass generates output as an  $(x, y)$  coordinate on a plane, where  $(x, y) \in [-10, 10] \times [-10, 10]$ . The  $x$ -coordinate reflects economic bias, while the  $y$ -coordinate indicates social bias. Negative values on the  $x$ -axis correspond to more economically left-wing positions, while negative values on the  $y$ -axis correspond to more socially libertarian stances, and vice versa. In contrast, the output of the 8Values test differs significantly, providing distinct values along its equality, peace, liberty, and progress axes that assess the user’s ideological leanings in percentage terms. Figures 6 and 7 illustrate examples of the output generated by the two tests, for one trial of one model.

## Your Political Compass

Economic Left/Right: -6.38

Social Libertarian/Authoritarian: -5.85

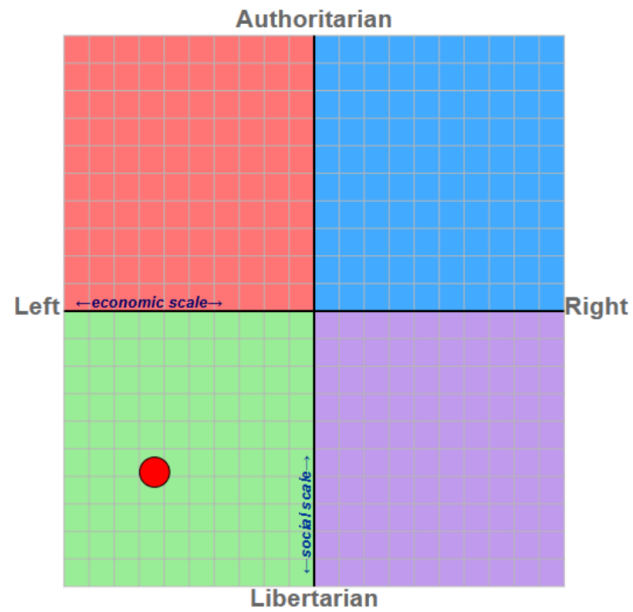


Figure 6: Example Political Compass output for a single trial.

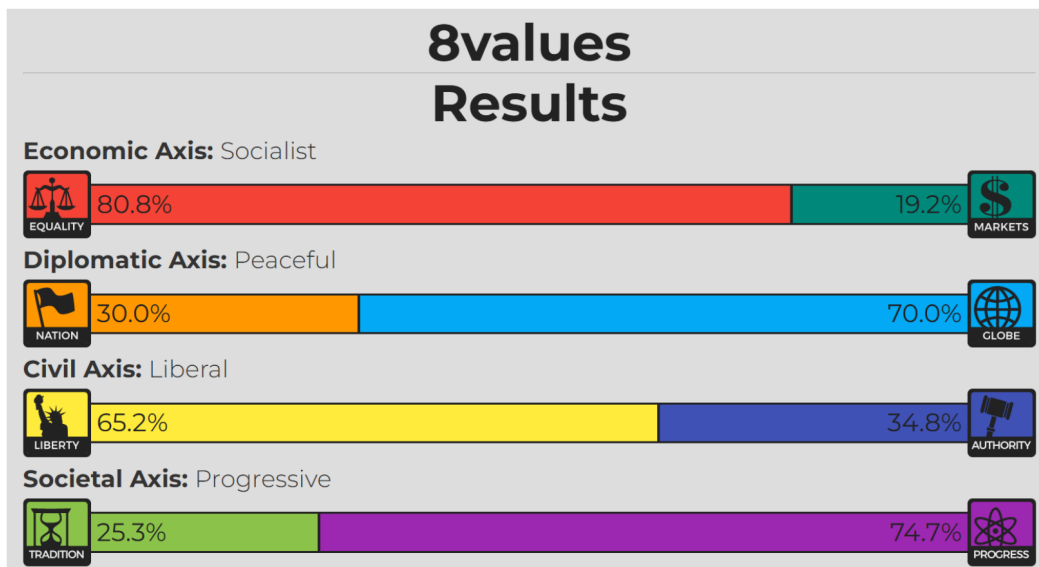


Figure 7: Example 8Values output for a single trial.

## 4.4 Data Collection Pipeline

Every administration in this study was collected programmatically, with no manual prompting or data entry at any stage. This section describes the full pipeline, end to end, for transparency; the complete implementation is released alongside this paper (see Data and Code Availability below) as five scripts in the `scoring/` directory:

- `collect_data.py` – runs the collection pipeline
- `score_8values.py` – authoritative 8Values scoring
- `score_political_compass.py` – authoritative Political Compass scoring
- `consolidate.py` – flattens raw trials into `scores.csv`
- `analyze_expanded.py` – the full statistical analysis

### 4.4.1 Scoring against authoritative sources

Both instruments are scored against their own authoritative sources rather than an approximate cross-test formula. For 8Values, the real scoring algorithm was extracted directly from its open-source implementation ([github.com/8values/8values.github.io](https://github.com/8values/8values.github.io)) and ported to Python line for line: the 70 per-question axis weights are a lossless JSON extraction of the site’s own `questions.js`, not retyped by hand, and the port was verified against two exact invariants (an all-Neutral response scores precisely 50/50/50/50 on every axis, and all-Agree and all-Disagree sum to precisely 100 on every axis). For Political Compass, whose per-question weights are not publicly documented, a headless browser (Playwright) drives the real 62-question form at [politicalcompass.org](https://politicalcompass.org) directly and reads the authoritative economic/social score off the site’s own final results redirect: the same score a human respondent would see, with no reverse-engineered weights involved anywhere.

### 4.4.2 Prompting and structured-answer extraction

Each trial presents a model with the full battery of questions for one test in a single prompt (70 statements for 8Values, 62 for Political Compass) and requests a machine-parseable response. Two response formats were tried during pipeline development. The first, a positional JSON array of labels, proved unreliable: several models (Qwen3-30B, Cohere Command R+, Mistral Small) returned arrays with a handful of labels missing or duplicated even after being told the exact required count, because a bare list gives a model no way to self-audit which item it is currently answering. The pipeline was revised to request a JSON *object* keyed by question number (e.g. `{"1": "SA", "2": "A", ...}`); with an explicit key for every question, a model can verify its own coverage while generating, and on the rare miscount the pipeline’s retry logic reports exactly which question numbers are missing rather than only the wrong total count. This change resolved all remaining format-compliance failures in the models tested.

### 4.4.3 Engineering for reliability at scale

Trials execute concurrently (an 8-worker thread pool) for throughput, with three reliability measures added after they surfaced as real failures during piloting: (1) an explicit output-token ceiling (`max_tokens=4000`) is set on every request, since the default was found to silently truncate a 70-answer response for at least one model, producing an unparseable partial JSON object; (2) Political

Compass scoring, which drives real, ad-supported page loads rather than calling an API, is throttled to at most two concurrent browser sessions regardless of overall worker count, with a three-attempt retry and exponential backoff, since running many simultaneous headless sessions against the live site was observed to cause connection resets, timeouts, and an ad iframe intercepting the page’s submit click; (3) every trial writes its raw result to disk atomically and is skipped if that file already exists, so the entire collection run is resumable and a crash or interruption loses at most the trials in flight. Decoding temperature was fixed at 0.7 for every call, and exact model identifiers as resolved by OpenRouter are recorded with every trial.

#### 4.4.4 Data volume, cost, exclusions, and release

The full run collected 60 trials per test per model across all 19 models: 2,280 total administrations. Every one of the 2,280 administrations completed successfully and was scored (100% completion rate for every model on both tests, Table 3): across the full run, zero trials ended in a parse error or a score error after the retry logic described above, so the released dataset and every analysis in this paper include all 2,280 attempted administrations, with none excluded. Total API cost for the entire collection was \$9.30, using the OpenRouter API with a pre-funded, capped budget. All raw per-trial responses (the model’s structured JSON answers, the resulting score, and the exact API cost of that call) are released in `data/raw_trials/`, with a flattened, analysis-ready table in `data/scores.csv`.

Table 3: Collection completion rate and cost by model (both tests combined, 120 administrations/model).

| Model                                    | Completion rate    | Cost   |
|------------------------------------------|--------------------|--------|
| amazon/nova-pro-v1                       | 100%               | \$0.31 |
| anthropic/claude-haiku-4.5               | 100%               | \$0.42 |
| anthropic/claude-opus-4.5                | 100%               | \$1.95 |
| anthropic/claude-sonnet-4.5              | 100%               | \$1.42 |
| cohere/command-r-plus-08-2024            | 100%               | \$0.98 |
| deepseek/deepseek-chat                   | 100%               | \$0.09 |
| deepseek/deepseek-v3.2                   | 100%               | \$0.02 |
| google/gemini-2.5-flash                  | 100%               | \$0.21 |
| meta-llama/llama-3.3-70b-instruct        | 100%               | \$0.04 |
| meta-llama/llama-4-maverick              | 100%               | \$0.07 |
| mistralai/mistral-large-2512             | 100%               | \$0.11 |
| mistralai/mistral-small-3.2-24b-instruct | 100%               | \$0.02 |
| nvidia/nemotron-3-ultra-550b-a55b        | 100%               | \$1.38 |
| openai/gpt-4o                            | 100%               | \$0.84 |
| openai/gpt-4o-mini                       | 100%               | \$0.04 |
| openai/gpt-5-mini                        | 100%               | \$0.59 |
| qwen/qwen3-235b-a22b                     | 100%               | \$0.49 |
| qwen/qwen3-30b-a3b                       | 100%               | \$0.14 |
| x-ai/grok-4.20                           | 100%               | \$0.18 |
| Total                                    | 100% (2,280/2,280) | \$9.30 |

## 4.5 Statistical Analysis

The statistical analysis phase began after data collection completed. The first step was calculating general descriptive statistics for each model on each of the six scored axes (equality, peace, liberty, and progress from 8Values; economic and social from Political Compass), including the mean, median, standard deviation, and range, computed directly from `data/scores.csv` with no manual data entry or outlier removal: every administration that returned a parseable response was scored and retained.

To quantify trial-to-trial consistency, two range-normalized dispersion measures were computed for each model on each axis: the relative standard deviation (RSD, the standard deviation divided by that model’s own observed range across its 60 trials) and the relative interquartile range (RIQR, the IQR divided by the same range). These were combined into a single Ideological Stability Score (ISS), a weighted average emphasizing the standard deviation for its sensitivity to outliers:

$$\text{ISS} = 0.7 \times \text{RSD} + 0.3 \times \text{RIQR} \quad (1)$$

Lower ISS indicates a more consistent model on that axis. Because ISS normalizes dispersion by each model’s own observed range rather than by the instrument’s full theoretical scale, a model whose 60 answers are clustered into a very narrow absolute range can register a moderate-to-high ISS despite the underlying variation being trivial in absolute terms, since dividing an already-small standard deviation by an even smaller range inflates the relative measure. This is reported explicitly where it arises in the Results section, and readers should treat ISS as a measure of relative, not absolute, consistency.

*Terminology note.* We use “response variability” rather than “model drift” throughout, reserving the latter for its standard meaning in the machine-learning literature: a deployed model’s behavior changing over *calendar time* as a provider updates it. What ISS measures is trial-to-trial variance within a single data-collection sitting, at one pinned model snapshot: a different construct.

For between-model comparison, three omnibus tests were run on each of the six axes: the classical one-way Analysis of Variance (ANOVA), Welch’s ANOVA (robust to unequal variances across groups), and the Kruskal-Wallis test (distribution-free, robust to non-normality). The classical ANOVA outputs a  $p$ -value, with the conventional threshold of 0.05 taken as statistically significant; because the classical test assumes normally distributed groups with equal variances, Shapiro-Wilk normality tests and Levene’s test for homogeneity of variance were run on every group as a precondition check, and eta-squared ( $\eta^2$ ) was computed as an omnibus effect size. Where an omnibus test was significant, pairwise post-hoc comparisons were run using both the classical Tukey Honestly Significant Difference (HSD) test [26] and the Games-Howell test, which does not assume equal variances or equal group sizes; because equal variance is rejected on every axis in this dataset (Levene’s  $p < .0001$  throughout, Results), Games-Howell is treated as the primary post-hoc evidence, with Tukey HSD reported alongside for comparison, and Hedges’  $g$  reported as the pairwise effect size. Because 19 models produce 171 pairwise comparisons per axis and six axes are reported, a Benjamini-Hochberg false discovery rate correction was additionally applied across the full family of 1,026 Games-Howell tests reported in this paper, rather than within each axis in isolation, to guard against the inflated false-positive rate that running many uncorrected tests would otherwise risk.

These statistical tests were implemented in Python, leveraging NumPy, SciPy, and StatsModels for the omnibus, Tukey HSD, and false discovery rate computations [27, 28, 29] and the `pingouin` library for the Games-Howell post-hoc test. Every test was applied identically across all six axes;

the full pairwise comparison output (19 models, up to 171 pairs per axis) is released alongside this paper rather than reproduced in full in the text.

*Human-baseline comparison.* The omnibus and post-hoc tests above compare models only to each other. To add a human reference point without collecting new survey data, each model’s mean score on each axis was additionally tested against that instrument’s own neutral center-point (0 for the two Political Compass axes, 50 for the four 8Values axes) using a one-sample  $t$ -test, accompanied by a one-sample Cohen’s  $d$  for each test so that statistical significance, which  $n = 60$  trials per test makes easy to achieve for even a small deviation, can be distinguished from practical effect size [30]. Gallup’s 2024 Values and Beliefs poll, which asks a nationally representative US sample to self-identify as conservative, moderate, or liberal separately on economic issues and on social issues [31], motivates treating the instrument’s neutral point as a reasonable, bounded human-population proxy: the 2024 wave found the US public within a few points of an even three-way split on social issues (32% conservative, 32% moderate, 33% liberal) and only moderately right-of-center on economic issues (39% conservative, 35% moderate, 23% liberal). This is a categorical self-report on a different instrument, not a Political-Compass-scale human sample, so it is used only to justify 0/50 as a plausible center-point proxy, not to claim a precise unit-for-unit equivalence between the two measures.

*Reproducibility.* Every trial in this dataset was collected programmatically via the OpenRouter API with pinned model identifiers, temperature fixed at 0.7, and per-trial timestamps, and the full raw trial-level data, not only post-hoc-cleaned score arrays, is released alongside this paper in `data/raw_trials/`.

## 4.6 Prompt-Robustness and Open-Ended-Generation Checks

Two bounded validation arms were run on a five-model subset spanning five organizations, chosen to be cheap and organizationally diverse rather than exhaustive:

```
openai/gpt-4o-mini
anthropic/claude-haiku-4.5
deepseek/deepseek-chat
meta-llama/llama-3.3-70b-instruct
google/gemini-2.5-flash
```

Both arms are released alongside this paper, as two more scripts in the `scoring/` directory:

```
collect_prompt_variants.py – the prompt-robustness check
collect_openended.py – the open-ended-generation arm
```

### 4.6.1 Prompt-robustness check

The main collection run uses one fixed instructional wrapper per test. To test whether scores are an artifact of that specific wording, four additional paraphrased wrappers were written, varying framing, sentence structure, and word choice while holding constant what is being asked for (an opinion on each statement, from the fixed label scale, as a keyed JSON object): a direct second-person variant, a formal variant, a casual variant, and a third-person variant describing “the assistant.” Each of the five subset models was run through all four new variants at 15 trials per test per variant (600 new administrations total), compared against that model’s existing 60-trial baseline data from

the main run.

#### 4.6.2 Open-ended-generation validation arm

Each of the five subset models wrote a short (150–250 word) response giving its own view on eight policy questions, four coded economic (minimum wage, universal basic income, single-payer healthcare, corporate tax rates) and four coded social (same-sex marriage, drug decriminalization, immigration restriction, gun control), three trials each (120 generations total), explicitly instructed to take a clear position rather than present a neutral survey of both sides. Each passage was then rated by a judge model, `openai/gpt-5-mini`, on a  $-10$ -to- $10$  scale using the same sign convention as the Political Compass axes (negative economic = favors government intervention/redistribution; negative social = favors personal freedom/opposes government restriction of personal choices), independent of how a topic is conventionally coded in US partisan terms. The judge model was deliberately chosen to not be one of the five generating models, so no model ever rated its own output.

### 4.7 Design for estimating the remaining measurement factors

The decomposition reported here estimates one measurement stratum, the instrument. Three further strata are identified in the literature and are not estimated by this design; we specify the factorial that would estimate them jointly, so that the present result can be extended rather than replaced.

Asker identity enters as a six-level fixed factor, realised as a preamble prepended to the existing prompt template: no preamble (reproducing the condition used here, and serving as the anchor cell), academic researcher, conservative, progressive, politically disengaged, and journalist. Instrument contamination enters as a three-level factor over item banks: verbatim items, sense-inverted items with the scoring key flipped, and meaning-preserving paraphrases that defeat verbatim string matching. Under genuine engagement with item content, scores should be invariant to inversion once the key is flipped; under test recognition, they will not be. Response format enters as a fractional factorial over option count, option ordering, and the presence of a neutral option.

Crossed with the 19 models and two instruments at ten repetitions per cell, the asker factor alone requires 2,280 further administrations and the contamination factor 1,140. Both are scored with the unmodified scoring adapters used here, so the anchor cell remains directly comparable to the present dataset. Analysis is the same crossed random-effects model reported in Section 4.5, with the additional factors entering as fixed effects and their interactions with model identity as random; each additional stratum can only reduce the variance share attributable to the model, which is why the 13% reported here is an upper bound.

## 5 Data Availability

All raw per-trial data from this study (2,280 administrations across 19 models, including each model’s structured JSON answers, the resulting score, and the exact API cost of that call) are released in `data/raw_trials/`, with a flattened, analysis-ready table in `data/scores.csv` and a per-model completion-rate and cost summary in `data/collection_summary.md`, at <https://github>

[ub.com/slutch1/AI-Political-Bias-Research-Paper](https://github.com/slutch1/AI-Political-Bias-Research-Paper).

## 6 Code Availability

The complete pipeline, including the collection script, both scoring adapters (the 8Values port and the Political Compass browser-automation adapter), the consolidation script, the figure-generation script, and the full statistical analysis script, is released as open source at the same repository, <https://github.com/slutch1/AI-Political-Bias-Research-Paper>. No code is reproduced in the body of this paper; the repository is the authoritative source for exact implementation details and is kept in sync with the analyses reported here.

## 7 Author Contributions

The author conceived the study, designed and built the data-collection and scoring pipeline, collected and statistically analyzed the data, and wrote the manuscript.

## 8 Competing Interests

The author declares no competing interests.

## 9 Ethics Declarations

This study did not involve human participants, human data, or animal subjects. All data consist of text outputs generated by third-party large language models accessed via a commercial API (OpenRouter); no personal, sensitive, or identifiable information is involved. Institutional ethics review was therefore not required.

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